LEARNER GUIDE

What can I now explain about a line?

Straight Lines and Connected Representations. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

This is the last lesson of the unit — a chance to draw everything together and see how much you now understand. You will look at one line told four ways at once (equation, graph, table and story), take a mixed challenge that touches every skill from the unit, and then meet the big idea that rounds it all off: a straight line is a model, and every model has its limits. Real batteries stop at 0%, real candles burn out, real plants don't grow forever. The question you are exploring is: what do you understand about a straight line — and where does a line stop being a good model?

What you are learning to notice

- That the equation, graph, table and story are four views of the same line — find m and c and you can move between them.
- That the whole unit rests on two numbers (m and c) and a few relationships (parallel, crossing, converting).
- That a straight line is a model of reality, not reality itself.
- That a good modeller can write the line AND say where it stops making sense.

Moving through it, one step at a time

In The whole story of a line, check that the equation, graph, table and story all describe the same line. In the Mastery Challenge, work through a mixed run — read a gradient, read an intercept, write an equation, spot a parallel line, find a break-even, do a conversion — and see how much comes back easily. In Where does the model break?, decide where each real situation stops following its line. Then, in A line that tells a human story, model a real situation and name where it runs out. There is no rush, and it is completely fine to reopen an earlier lesson if an idea needs a second look — that is good learning, not falling behind.

A gentle reminder
You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the whole unit — reading and writing y = mx + c, parallel lines, intersections, and conversion graphs (open any earlier lesson if a part feels shaky)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.

- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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